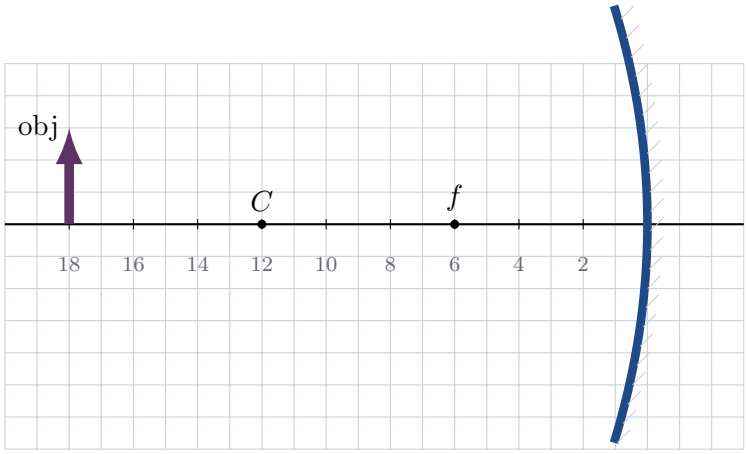
1. A concave mirror has a radius of curvature $R = 12$ cm. A 3 cm tall object stands 18 cm in front of it. On the grid, one small square is 1 cm.



A. Draw all **three** principal rays from the tip of the object. Mark the image.

B. From your drawing, estimate d_i, h_i and M.

$d_i \approx$ _____ $h_i \approx$ _____ $M \approx$ _____

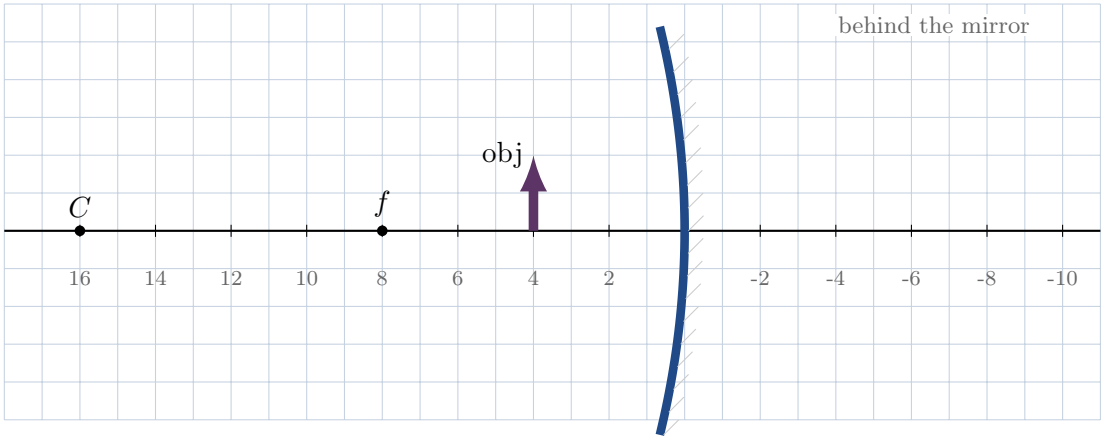
C. Is the image real or virtual? Upright or inverted? Bigger or smaller?

2. A convex mirror (like the passenger mirror on a car) has $|R| = 30$ cm. A 6 cm tall object is 20 cm in front of it.

A. Find f (watch the sign), then use the mirror equation to find d_i.

B. Find M and h_i. Is the image real or virtual? Upright or inverted?

3. A concave makeup mirror has $R = 16$ cm. You hold a 2 cm tall object 4 cm in front of it, **closer than** f. On the grid, one small square is 1 cm.



A. Draw ray 1 (parallel → through f) and ray 2 (on the line from f through the tip → parallel). The reflected rays spread apart, so extend them **behind** the mirror with dashed lines and mark where they meet.

B. Where is the image? Is it real or virtual? Upright or inverted? Bigger or smaller?

C. Check with the mirror equation: find d_i and M. Do they match your drawing?